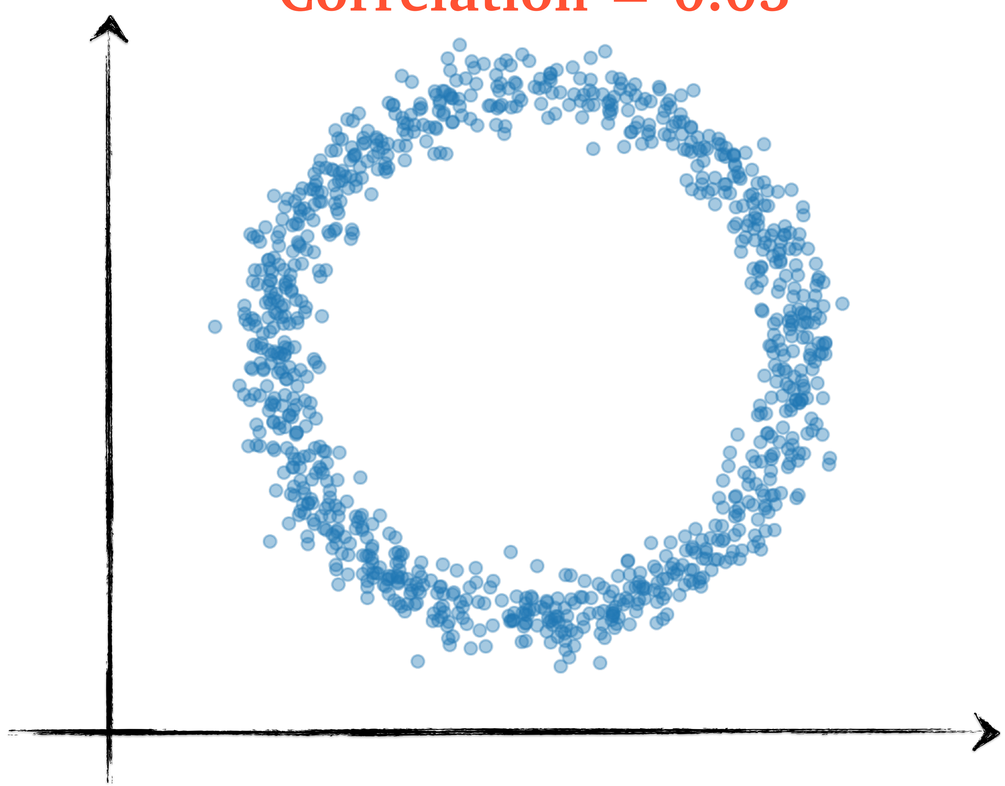
Why Linear Statistical Dependence Is Not Enough ?

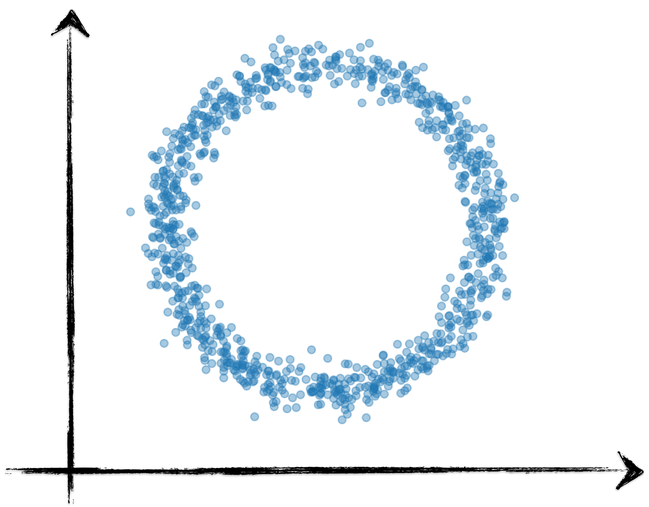
$$\text{correlation} = \frac{\mathbf{Cov}(X, S)}{\sigma_X \, \sigma_S} = \frac{\mathbb{E}\left[(X - \mu_X)(S - \mu_S)\right]}{\sigma_X \, \sigma_S}$$

Correlation = 0.03



Independence $\rightarrow$ Correlation = 0

Correlation = 0 $\not\rightarrow$ Independence

How Do We Capture Nonlinear Statistical Dependency ?

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